

Karl-Alexandre Michaud

karlthedeveloper.tech | linkedin.com/in/karlmichaud | github.com/Karl-Michaud

EDUCATION

University of Toronto

Toronto, Ontario, CA

Honours Bachelor of Science, Computer Science (Focus in Computer Systems), Mathematics Sep 2023 – Jun 2027

- **Courses:** Software Design, Software Engineering, Operating Systems, Parallel Programming

EXPERIENCE

Software Developer

Sep 2025 – Present

saycamel.com

Toronto, Ontario, CA

- Built a **social platform** with **850+** registered users and **4.3k+** monthly visitors, connecting students to **1400+** UofT clubs and **240+** weekly events using **React**, **JavaScript**, and **Supabase database**
- Led a team of 3 to develop the club dashboard and **collaborated** with cross-functional teams to design and deploy **end-to-end features**, supporting **100+** prospective clubs and **increasing engagement by 28%**

Software Developer (Markus, PythonTA)

May 2025 – Present

University of Toronto, Department of Computer Science

Toronto, Ontario, CA

- Engineered and maintained **MarkUs**, a scalable production-grade web service supporting **10k+** daily users, using **Ruby on Rails**, **React**, **PostgreSQL**, and **Docker** with **CI/CD pipelines** for reliable and secure deployment
- Developed and maintained **backend services and technical documentation** for **PythonTA**, a static code analysis tool used by **35k+** students in UofT courses since 2016, leveraging **Python** and **Astroid**
- Built **automated test frameworks** using **Pytest**, **Rspec**, and **Jest** within **CI/CD GitHub Actions pipelines**, enabling continuous integration testing for production systems

PROJECTS

CExpress | C, Shell, Make, Homebrew, Git

karl-michaud.github.io/CExpress/

- Created **CExpress**, a lightweight, high-performance **HTTP server / microservice framework** for **C** that enables rapid prototyping and development of **RESTful APIs** with minimal overhead
- Implemented key features such as **dynamic routing**, **HTTP/1.1 request handling**, and low-level **TCP socket programming**, showcasing expertise in **network programming** with an **optimized memory footprint**

AI Video Generator | C, Python, Shell, FFmpeg, Deepseek API, Whisper, Git

www.youtube.com/watch

- Engineered an **AI video generation pipeline** in **C**, automating script generation, TTS synthesis, subtitle burning, and media merging using **FFmpeg**, **Whisper**, and integrating the **DeepSeek API**
- Architected a **multi-process pipeline** of 5 standalone CLI programs communicating through **Unix pipes**

Recyclobot – DeltaHacks 11 | Swift, Python, Firebase, OpenAI API, Git

devpost.com/software/recyclobot

- Built an AI-driven **waste management system** with **real-time object detection** and sustainability insights
- Developed **back-end microservices and data pipelines**, integrating OpenAI API and Firebase database
- Delivered a functional **iOS app + hardware prototype**, ranking **top 12 overall** and **top 3 environmental**

NorthBlue Network | Next.js, TypeScript, Supabase, Git

www.northblue.network

- Built a scalable **networking platform** connecting UofT builders, creators, and founders, generating **10K+** impressions in 24 hours and attracting users including **4 VC investors** and notable startup founders
- Implemented authentication, profile discovery, and real-time updates using **Next.js**, **TypeScript**, and **Supabase**

LEADERSHIP AND INVOLVEMENT

Event Coordinator

Aug 2024 – May 2025

University of Toronto, STEP UofT Student Club

Toronto, Ontario, CA

- Led **event planning** during the club's founding year, organizing **5 events** with **15+** attendees each
- Onboarded **25+ new members**, driving club growth through strategic **communication and collaboration**
- Secured **5 UTSL Club Award nominations** in our inaugural year by improving event logistics

TECHNICAL SKILLS

Languages: Python, JavaScript, C, Java, Ruby, Swift, HTML, CSS, SQL, Shell, MIPS Assembly, Makefile

Frameworks / Libraries: Rails, Node.js, Next.js, SwiftUI, Maven, React, Pytest, JUnit, Astroid, Rspec

Developer Tools: Git, GitHub, Docker, CI/CD (GitHub Actions), PostgreSQL, Supabase, Firebase, Linux